

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) — Semester IV Examination, 2013-14

Paper No. MTB-402: Mathematical Methods

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory.)

- (a) Using the definition of the Laplace transform, find $\mathcal{L}\{\sin at\}$. [2]
- (b) Prove that the Laplace transformation is linear. [2]
- (c) Using the convolution theorem, find the value of $\mathcal{L}^{-1}\left\{\frac{1}{s(s+a)}\right\}$. [3]
- (d) Find the Laplace transform of $f(t)$, where $f(t) = e^t$ for $0 \leq t < 2\pi$ and $f(t+2\pi) = f(t)$. [3]
- (e) Write the function whose Laplace transform is 1. [1]
- (f) If $\mathcal{L}^{-1}\{F(s)\} = f(t)$, then prove that $\mathcal{L}^{-1}\{F(as)\} = \frac{1}{a}f\left(\frac{t}{a}\right)$. [2]
- (g) If $f(x)$ has the Fourier coefficients a_n, b_n , show that $kf(x)$, where k is a constant, has the Fourier coefficients ka_n, kb_n . [2]
- (h) If the Fourier cosine integral of $f(x)$ is given by $f(x) = \int_0^\infty A(\omega) \cos(\omega x) d\omega$, prove that $-\frac{d^2}{dx^2}f(x) = \int_0^\infty \omega^2 A(\omega) \cos(\omega x) d\omega$, where $A(\omega) = \frac{2}{\pi} \int_0^\infty f(x) \cos(\omega x) dx$. [2]
- (i) Define the n -th order proximity of two curves. [2]
- (j) Test for an extremum for the functional $I[y] = \int_0^1 (y'^2 + y^2 - 2y'^2 y') dx$, $y(0) = 1$, $y'(0) = 2$. [3]

Question 2.

- (a) If $f(t)$ is a continuous function except for an ordinary discontinuity at $t = a$ ($a > 0$), then show that:

$$\mathcal{L}\left\{\frac{d}{dt}f(t)\right\} = s\mathcal{L}\{f(t)\} - f(0^+)$$

[6]

(b) Express $f(t)$ in terms of unit step functions and hence find its Laplace transform, where:

$$f(t) = \begin{cases} 0, & 0 < t < 1 \\ t - 1, & 1 < t < 2 \\ 1, & t > 2 \end{cases}$$

[6]

Question 3.

(a) State and prove the existence theorem for the Laplace transform.

[6]

(b) Using Laplace transforms, solve the differential equation:

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - y = x, \quad y(0) = 0, \quad y'(0) = 1$$

[6]

Question 4.

(a) Find $\mathcal{L}^{-1}\left\{\frac{\ln(s+a)}{s+b}\right\}$.

[6]

(b) Find the Fourier cosine series expansion of the periodic function of period $2P$, defined by $f(t) = \sin\left(\frac{\pi t}{P}\right)$ for $0 < t < P$.

[6]

Question 5.

(a) Using Laplace transforms, solve the integral equation:

$$y(t) = \sin 2t + \int_0^t y(x) \sin 2(t-x) dx$$

[6]

(b) Find the Fourier integral representation of the function:

$$f(x) = \begin{cases} 0, & x < 0 \\ 1, & 0 < x < 1 \\ 0, & x > 1 \end{cases}$$

Hence find the value of $\int_0^\infty \frac{\sin \omega}{\omega} d\omega$.

[6]

Question 6.

- (a) Find $\mathcal{L}^{-1}\left\{\frac{5s+3}{(s-1)(s^2+2s+5)}\right\}$. [6]
- (b) Using the calculus of variations, find the shortest distance between the point $(1, 0)$ and the ellipse $4x^2 + 9y^2 = 36$. [6]
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Question 7.

- (a) Find the extremal for which the functional $I = \int_0^1 y'^2 dx$, $y(0) = 0$, $y(1) = 1$ is an extremum, subject to the isoperimetric condition $\int_0^1 y dx = \frac{1}{6}$. [6]
- (b) Find the extremal for the functional:

$$I = \int_0^{\pi/2} (y''^2 - y^2 + x^2) dx, \quad y(0) = 0, \quad y'(0) = 1, \quad y\left(\frac{\pi}{2}\right) = y'\left(\frac{\pi}{2}\right) = \frac{1}{4}$$

[6]